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Optimizing Asset Replacement Decisions Through Integrated 360° Insights

Hany Elkhazhi, Muzna Abdulla Alnoaman, and Basma Ali Ahmed, ADNOC Offshore, Abu Dhabi, United Arab Emirates

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Abstract

Objectives/Scope: Upper Zakum field is a significant offshore site where ADNOC Offshore invests heavily to bringing development target to reality. UZ consists of multiple offshore wellhead platforms, satellites, super complex and four artificial islands. This mega offshore field development plan consists of increasing its target production to One and half Million barrels per day by 2031. The UZ registered assets number in the hundreds of thousands, possibly will be doubled after ongoing development projects are completed. These assets vary in age, including both older and newer ones, representing different stages in the asset lifecycle. These assets record a massive volume of information during their life cycle including the asset identification data, technical specification, financial and procurement records, operational history, Maintenance and modification data; in addition to the track records of asset performance where all these data are contributing to the remnant life assessment. Utilization data including timeline and volumes whether for production or injection is also contributing to selecting the optimum mitigation action towards the deteriorated asset by comparing the mitigation expected life extension with the utilization plan as this will lead to huge cost optimization. The integration of these comprehensive data is critical for informed asset replacement decisions. Without a unified 360° view, organizations risk making expensive, reactive choices based on fragmented insights. Integrated data enables predictive analytics (e.g., forecasting equipment failure), accurate total-cost-of-ownership (TCO) calculations, and scenario modeling (e.g., comparing replacement vs. life extension). Siloed data, in contrast, leads to blind spots—such as underestimating revenue impacts or missing compliance deadlines. In asset-intensive industries, this integration isn't just beneficial; it's a strategic imperative to maximize up time, optimize spending, and mitigate risks. Meanwhile, in an oil field with intensive asset locations utilization of interactive maps will facilitate further the data integration and allow users to explore asset locations detailed information visually once clicking that location on the map.

Methods, Procedures, Process: The Asset Replacement Dashboard is a Decision Support Tool that integrates asset information from various systems, such as SAP system including fixed asset register(master data) as well as maintenance data, asset remnant life recorded in Passport and Reliability systems, replacement projects budget; in addition to the asset utilization plan represented by the production and injection forecasts.

Results, Observations, Conclusions: This dashboard resulted in establishing a robust, integrated platform that consolidates all critical asset-related information into a centralized dashboard enabling data-driven decisions that help avoid premature replacements and ensure optimal asset utilization. Realized cost optimization has been achieved as a result of several optimization for replacement projects scopes that were recorded such as avoidance of pipelines and helipad replacement.

Introduction

The Upper Zakum (UZ) field is one of ADNOC most strategic offshore assets, representing a cornerstone of the company's long-term development vision. Spanning four distinct geographical zones, each comprising multiple offshore wellhead platforms, satellite or super complex, and artificial island, UZ has undergone successive waves of development over the past 50 years. This evolution is set to continue, with ADNOC targeting a production capacity of 1.5 million barrels per day by 2031. To achieve this ambitious plan is in place to invest billions of dollars in facilities, pipelines, islands, platforms, and other assets. Maximizing the value of such brownfield infrastructure, while maintaining safe, reliable, sustainable operations is a strategic imperative for ADNOC offshore. A robust Asset Life Cycle Management (ALM) framework is employed to manage this imperative. This paper describes an integrated digital platform that underpins this framework.

The UZ registered assets number in the hundreds of thousands, possibly will be doubled after ongoing development projects are completed. These assets are at diverse phases in their life cycles and record a massive volume of information during their life cycle including the financial and procurement records, operational history, maintenance and modification data; in addition to the track records of asset performance where all these data are contributing to the remnant life assessment. Utilization data including timeline and volumes whether for production or injection is also contributing to selecting the optimum mitigation action towards the deteriorated asset by comparing the mitigation expected life extension with the utilization plan as this will lead to huge cost optimization. The integration of these comprehensive data is critical for informed asset replacement decisions. Without a unified 360° view, organizations risk making expensive, inaccurate choices based on fragmented insights. Integrated data enables predictive analytics (e.g., forecasting equipment failure), accurate total-cost-of-ownership (TCO) calculations, and scenario modeling (e.g., comparing replacement vs. life extension). Siloed data, in contrast, leads to blind spots such as underestimating revenue impacts or missing compliance deadlines. In asset-intensive oil fields, this integration is not only beneficial; it's also necessary for maximizing the asset uptime, minimizing costs, and lowering risks. Meanwhile, the use of interactive maps will facilitate distribution the massive volume of data, and allow users to explore asset locations detailed information visually once clicking that location on the map.

The Asset Replacement Dashboard is a Decision Support Tool that integrates asset information from various systems, such as SAP system including fixed asset register (master data) as well as maintenance data, asset remnant life recorded in Passport and Reliability systems, replacement projects budget; in addition to the asset utilization plan represented by the production and injection forecasts.

Objective

The primary goal of the Asset Replacement and Life Extension Plan is to maintain the continuous, reliable, and safe operation of ADNOC Offshore's (AOF) assets. This is achieved by systematically identifying and replacing assets that have reached the end of their useful life or no longer meet performance standards. The anticipated benefits include improved reliability, reduced operational risks, and compliance with safety regulations (ADNOC Offshore Guideline for Asset Replacement Plan).

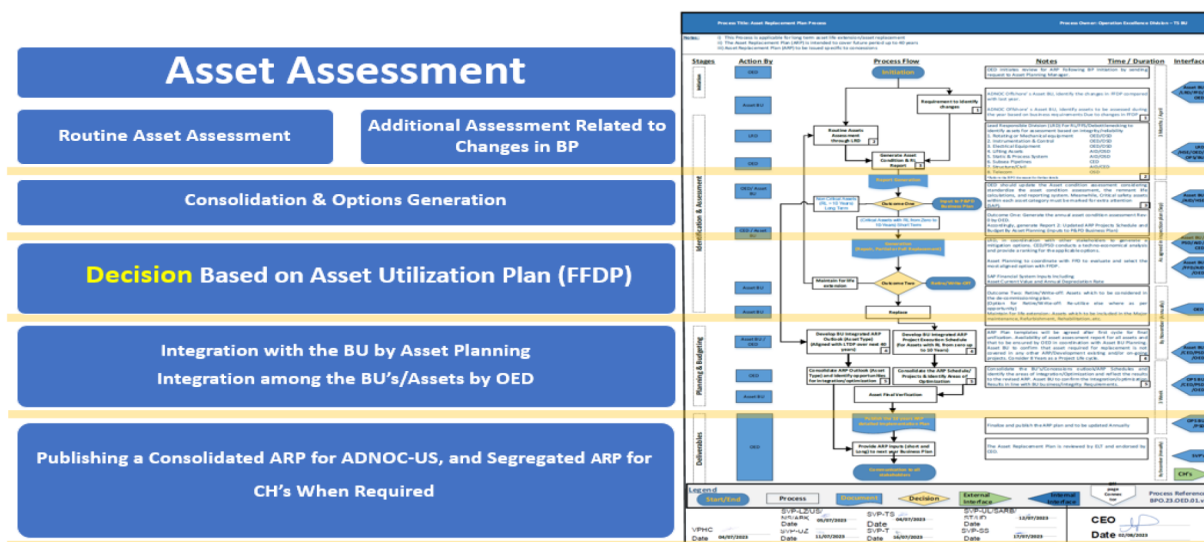
Whereas the primary objective of the Asset Replacement Dashboard is to enable strategic, data-driven decision-making for asset replacement across Upper Zakum field. By integrating and visualizing diverse asset-related datasets including asset location, operational status, work orders, wells performance

(production/injection), depreciation, technical remnant life, and budget allocations into a unified digital platform, the tool empowers stakeholders to make informed decisions that maximize asset utilization and avoid premature replacements.

This centralized perspective promotes collaboration across functions, improves the accuracy of lifecycle planning, and aids in cost optimization by aligning technical evaluations with operational forecasts and financial constraints. The dashboard converts fragmented data into actionable insights, enabling predictive analytics, scenario modelling, and total cost of ownership (TCO) evaluations. These capabilities are especially critical for managing the vast and aging infrastructure of the Upper Zakum field in a sustainable and efficient manner.

Integrated Data Sources and Workflow for Asset Replacement Plan (ARP)

The Asset Replacement Plan (ARP) process comprises three key stages as shown in the below picture, designed to ensure strategic, timely, and cost-effective asset replacement across ADNOC OFFSHORE fields. The initial stage focuses on asset integrity and reliability performance, followed by remnant life reports that documented the remaining useful life of each asset. These reports are essential for ensuring business continuity and timely asset replacement. The second stage aims to evaluate available replacement options in alignment with the utilization plan. This plan is derived from the field development strategy, enabling the selection of the most optimal replacement option based on operational priorities and long-term field objectives. The third stage involves selecting the best execution strategy that aligns with ongoing development projects and phased production targets outlined in the business plan. This ensures that replacement activities are synchronized with broader field initiatives and resource mobilization efforts.



To manage the complexity and scale of asset replacement plan in the Upper Zakum field, ADNOC Offshore UZ Operation Planning Team developed the **Asset Replacement Decision Support Tool** that designed to integrate, visualize, and operationalize asset-related data across multiple systems and workflows.

This model consolidates data from key enterprise platforms, including:

- **Integrity and Reliability Systems:** Asset performance reports are used to determine the technical remnant life of each functional location. To ensure timely and consistent assessments, the SAP system is utilized to track and validate that each location maintains a remnant life exceeding the duration required to execute and complete its associated replacement project. This

approach ensures that replacement planning is aligned with asset health and operational readiness, minimizing the risk of premature interventions or project delays.

- **SAP:** The primary modules utilized include the Fixed Asset Register for managing master data, Maintenance Management for overseeing asset upkeep, Workflow Adjustment for process optimization, and Financial Management, which encompasses asset depreciation tracking and related financial records.
- **Production and Injection Forecasts and Wells Status per location:** The utilization plan for major locations—such as wellhead platforms (WHPTs), satellites, and Islands are directly tied to the forecasted production and injection profiles of each site. In parallel, individual well performance and its remnant life play a critical role in determining the expected operational lifespan of these locations, whether for oil production or water injection. Meanwhile, multiple organizational entities contribute to evaluating well performance, including the Field Development team, Reservoir Management, and Asset Integrity (Well Operations). To ensure alignment across these perspectives, consolidated well data view has been considered in the Decision Support Tool. This unified dataset enhances the accuracy of utilization forecasts and supports more informed decision-making regarding asset deployment and replacement timing.
- **Business Plan Forecasted Budget:** Project budget allocations are directly linked to the delivery schedule of field targets, whether they pertain to sustaining current production levels (base case, including ARP) or achieving growth targets aimed at increasing production capacity (development case). This alignment presents a strategic opportunity to merge asset replacement scopes with development scopes. By integrating these efforts, ADNOC Offshore has optimized replacement budgets by leveraging development-driven mobilization and infrastructure upgrades to address aging assets concurrently. This approach not only enhances cost efficiency but also ensures that replacement activities are executed in synergy with broader field development plans.

Case Study: Asset Replacement Plan Upper Zakum Field

The Upper Zakum field represents one of the most complex environments for asset lifecycle management within ADNOC Offshore, encompassing thousands of assets distributed across offshore platforms, artificial islands, and super complexes. This case study demonstrates how the Asset Replacement Decision Support Dashboard was applied to optimize asset replacement decisions in this high-density, high-value operational setting.

The implementation was particularly critical in light of ADNOC's aggressive development strategy, which emphasizes new facilities on islands while phasing out further development on WHPTs (Wellhead Platforms). This shift required close monitoring of WHPT facility remnant life, especially in relation to subsurface constraints that influence the utilization plan of each WHPT.

Context and Challenge

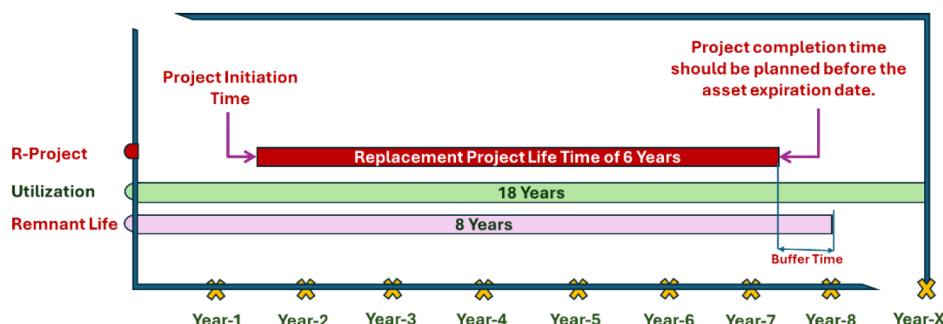
As part of ADNOC's strategic objective to increase production to 1.5 million barrels per day by 2031, the Upper Zakum field is undergoing extensive infrastructure expansion. With a mix of aging assets and newly commissioned equipment, planners faced the challenge of determining:

- Which assets required immediate replacement
- Which could be extended safely
- How to prioritize interventions across geographically dispersed locations

Traditional planning methods relied on manual reconciliation of data from platforms such as SAP, Passport, Reliability Systems, and production forecasts. This fragmented approach led to inefficiencies in

planning, reactive decision-making, and potential over-investment in replacements that could have been deferred or avoided.

Asset replacement decisions are governed by a structured framework that integrates technical, operational, and financial dimensions. A critical aspect of this framework is the comparison between an asset's remnant life and its utilization period, with the replacement project lifecycle typically defined as six years. This triad forms the cornerstone of a rational, risk-informed, and cost-effective asset management strategy.



The chart above illustrates a structured timeline for asset replacement planning, emphasizing the alignment between an asset's remaining life, its operational lifespan, and the time required to complete a replacement project. The asset's full operational lifespan (represented by the light green bar) extends over 18 years, indicating the total duration the asset is expected to support the field operations, while In contrast, the remaining useful life (shown in pink) is only 8 years, highlighting a critical window for initiating replacement planning. The replacement project duration (depicted in red) is set at six years, which reflects ADNOC Offshore's standard planning practice; where is the graph underscores the principle of replacement projects must be completed before the asset reaches the end of its useful life ensuring that capital expenditures are aligned with actual operational needs, avoiding premature replacements and minimizing stranded investments. However, In certain scenarios, particularly where an offshore wellhead platform is operating with only a few number of active wells and its connected pipeline exhibits a short remaining life, it becomes both strategically and economically prudent to accelerate the replacement of those wells through long-reach drilling from nearby artificial islands, where this approach aligns with the core asset replacement planning principle of avoiding investment in infrastructure with limited remaining life. It also circumvents the substantial capital and operational costs typically associated with pipeline replacement.

To address these challenges, ADNOC Offshore introduced the Asset Replacement Dashboard that integrates technical, operational, and financial datasets into a unified platform, the dashboard empowers stakeholders to make timely, risk-informed asset replacement decisions that align with strategic field development goals.

Dashboard Architecture and Key Functionalities

The dashboard is purposefully designed to support cross-functional workflows, offering a unified interface that fosters seamless collaboration among planners, engineers, and financial controllers. It promotes shared visibility and coordinates decision-making, ensuring alignment in asset planning and operational strategies. The interface presented in this paper was developed using Microsoft Power BI, capitalizing on its robust capabilities in data modelling, real-time visualization, and DAX-based logic processing.

As illustrated in the real dashboard image below, each numbered segment corresponds to a specific component described in the following section. These numbered references are intended to guide the reader through the dashboard's architecture, making it easier to connect visual elements with their respective functionalities and strategic purposes.

1. **Interactive Geographical Location Map:** Users can interactively explore asset locations by selecting mapped zones, which function as a visual search engine. Upon selecting a specific location, all related dashboards and visualizations are dynamically filtered to present context-specific data. This includes detailed asset metadata, associated wells, functional locations, the planned project scope, work order history, forecasted production and injection figures, and a comparison of average remnant life versus financial remnant life. This integrated view enables users to make informed decisions by focusing on the most relevant operational and financial insights for each asset.
2. **Function Location Master Table:** This comprehensive dataset integrates detailed functional location information sourced from the SAP EAM system with asset remnant life assessment results obtained from Asset Performance Management (APM) systems. The consolidation of operational and performance data provides a unified view that supports strategic decision-making in asset planning, lifecycle management, and maintenance prioritization.
3. **Workorder Analysis Graph:** This visualization leverages data sourced from the SAP EAM system to present the maintenance work order history by type, mapped to each asset's functional location. It serves as a valuable tool for refining the asset maintenance strategy, aligning it with operational requirements and assessment outcomes. Additionally, it supports the implementation of predictive maintenance practices wherever feasible, helping to optimize asset performance and reduce unplanned downtime.
4. **Financial and Technical Analysis:** This visualization compares two key dimensions of asset longevity. The technical remnant life data is sourced from the Asset Performance Management (APM) system, reflecting the operational condition and expected lifespan of each asset. In contrast, the financial remaining life data is derived from the Fixed Asset Register, which includes details such as Asset Book values, current valuations, and depreciation tracking. This comparative analysis supports more informed decision-making by aligning technical performance insights with financial planning and asset valuation.
5. **Production Injection Forecasts Graphs:** These graphs present indicative oil production and water injection forecasts, sourced from the annual business plan. They reflect the projected asset utilization timeline, offering insights into expected performance and operational planning across the asset lifecycle.
6. **Location Status and Assessment Status:** Pie charts are used to visualize the location status of assets, highlighting their operational state. This ensures that attention is directed exclusively toward active assets, excluding those marked as "Inactive" or "To Be Decommissioned.". In parallel, the Assessment Status metric is employed to compare assessed versus non-assessed assets across each location. This comparison supports the prioritization of future assessment plans, ensuring resources are allocated efficiently and strategically.
7. **Wells Status Table and Pie Chart:** These visuals present the monthly updated well status, sourced from field development databases such as Seabed and Digital 360, or through the Inactive Report. They play a critical role in supporting replacement decisions for WHPT functional locations by identifying the number of healthy wells associated with each tower. Additionally, they help determine the optimal timing for transitioning a tower's status from "Operational" to either "Inactive" or "To Be Decommissioned," ensuring alignment with field development strategies and asset utilization planning.
8. **Projects Budget Phasing:** This graph presents data extracted from the SAP Financial System during the development of the 5-Year Business Plan, categorized by Project Number. It enables stakeholders to monitor each project's scope, budget allocations, and timeline progression. Additionally, it provides a strategic opportunity to identify synergies between asset replacement. Scopes and development scopes, supporting more efficient planning and resource optimization.

9. **Cost Saving Graph:** This graph illustrates the financial impact of replacement avoidance, derived from the implementation of the Integrated Asset Replacement Plan Decision Support Dashboard. By consolidating technical, financial, and operational data into a unified decision-making framework, the dashboard enables more accurate identification of assets that can be extended or optimized rather than prematurely replaced. Ultimately, the graph serves as a key performance indicator for the dashboard's effectiveness in driving cost-efficient asset strategies aligned with ADNOC's broader goals for sustainability, production optimization, and value creation.
10. **Action Plan Table:** This table captures the action items arising from discussions on asset assessment results. It includes requests to accelerate assessments for specific functional locations, enabling a more comprehensive view of related areas and supporting informed decision-making for individual assets. Additionally, the table serves as a collaborative tool, facilitating communication across teams and promoting cross-functional alignment, thereby reducing siloed efforts and enhancing group coordination.



Moreover, the dashboard is equipped with advanced filtering capabilities including multi-level filters (F1) that enable users to quickly and intuitively navigate to specific functional locations. A category-based filter allows searching by asset category, including both the eight primary categories and their respective sub-categories. Additionally, users can filter by geographical area and major location, streamlining access to targeted assets—especially when combining category and location filters for precision.

To support assessment planning, a dedicated filter for Assessment Status highlights functional locations that have not yet been assessed within key operational zones.

Most importantly, a specialized filter has been added to identify functional locations within a specific remnant life range (F2), ensuring timely and strategic implementation of asset replacement or extension plans.

Decision Outcomes

Using the dashboard, Upper Zakum field was able to:

- **Avoid Unnecessary Replacements:** Several planned replacements, including pipelines and helipads, were deferred after analysis that showed sufficient remaining life aligned with the utilization plan which is led to lowering operational risks.
- **Prioritize High-Impact Assets:** Assets critical to production continuity were flagged for immediate action, while lower-risk items were scheduled for future review.
- **Improve Stakeholder Collaboration:** Engineering, operations, and finance teams used the dashboard to align on decisions, reducing delays and improving transparency.

Impact

This case study demonstrates how integrated data, and visual intelligence can transform asset replacement planning into digital form aligned with the Upstream 2.0 digital transformation vision. The dashboard enabled ADNOC Offshore to shift from reactive, fragmented decision-making to a proactive, strategic approach—maximizing asset value, minimizing unnecessary spending, and supporting long-term production goals.

The deployment of the ARP decision support tool across Upper Zakum asset portfolio yielded substantial operational and financial benefits. The tool was applied to a wide range of offshore infrastructure asset categories, including wellhead platform (WHPT), pipelines, cranes, helipads, and piping systems. The results including total cost avoidance of approximately USD 800 million for the following categories:

- Avoidance of replacement for four pipelines totalling ~14 km.
- Deferral of 44 offshore platform crane replacements.
- Exclusion of three WHPT helipads from replacement plan.
- Deferral of 16 WHPT piping systems.
- Optimization of maintenance scope for redundant and short-horizon WHPT's.

These avoided costs reflect not only direct capital expenditure savings but also reduced operational disruptions and improved lifecycle efficiency, and demonstrating the effectiveness of ARP decision support tool in transforming asset replacement planning from a reactive process into a proactive, data-driven strategy that supports long-term production goals and capital efficiency.

Conclusion and Future Outlook

The success of the ARP decision support tool lies in its ability to bridge the gap between technical asset data and strategic planning. Traditional asset replacement approaches often rely on age-based criteria or isolated condition assessments, leading to fragmented decisions and inefficient capital deployment. The ARP tool introduced a paradigm shift by integrating multidimensional data, visualizing asset risk and value, standardizing replacement logic, and enhancing collaboration across departments. It also supported ADNOC Offshore's digital transformation goals by embedding analytics into core planning processes.

The deployment of the Asset Replacement Dashboard as a decision support tool in the Upper Zakum field has marked a transformative step in ADNOC Offshore's approach to asset lifecycle management. By integrating diverse datasets into a unified, interactive platform, the dashboard has empowered stakeholders to make informed, strategic decisions that optimize asset utilization, reduce unnecessary replacements, and align capital investments with operational priorities.

The results achieved ranging from cost savings and improved planning accuracy to enhanced cross-functional collaboration demonstrate the value of data-driven decision support in managing complex offshore infrastructure. The dashboard has proven to be more than a visualization tool; it is a strategic enabler that supports ADNOC's long-term production goals and sustainability commitments.

Looking ahead, the evolution of this platform will be guided by the principles of **Project SHIFT**, that initiated by ADNOC OFFSHORE CEO is leading an initiative that aimed at transforming ADNOC Offshore's digital ecosystem. Project SHIFT envisions a future where systems are directly and automatically connected, minimizing manual intervention and ensuring **100% data accuracy**. This initiative will drive the next phase of integration linking SAP, Passport, Reliability Systems, and production forecasts through intelligent automation and real-time data synchronization.

Future enhancements to the Asset Replacement Dashboard will include:

- **Automated Data Validation:** Ensuring consistency and accuracy across systems without human reconciliation.
- **AI-Driven Recommendations:** Leveraging machine learning to suggest optimal replacement strategies based on historical patterns and predictive models.
- **Real-Time Alerts and Notifications:** Enabling proactive maintenance and risk mitigation.
- **Expanded Graphic Map Capabilities:** Providing deeper spatial insights into asset clusters, risk zones, and operational dependencies.

As ADNOC Offshore continues to scale its operations and modernize its infrastructure, the integration of Project SHIFT principles into asset management workflows will be critical. The Asset Replacement Dashboard stands as a foundational tool in this transformation bridging data, strategy, and execution to deliver operational excellence across the Upper Zakum field and beyond.

Acknowledgments

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Special thanks are due to the **Field Operations**, **Asset Integrity**, and **Operations Excellence** teams, as well as the **HSE Advisors** and **Subject Matter Experts**, whose active participation and domain expertise were vital throughout the ARP process. Their contributions ensured the tool's alignment with real-world operational needs and safety standards.

The authors also gratefully acknowledge the **ADNOC Offshore Digitalization Team** for their technical leadership in optimizing dashboard performance and securing seamless integration within the Company Portal. Their efforts have significantly enhanced the tool's accessibility, responsiveness, and analytical capabilities.

This work reflects the collective dedication of cross-functional teams committed to advancing ADNOC Offshore's digital transformation and asset management excellence.

Nomenclature

ARP	Asset Replacement Plan
TCO	total-cost-of-ownership.
HSE	Health, Safety, and Environment – organizational function concerned with workplace safety and risk.
AI	Artificial Intelligence – logic-based, automated evaluation embedded in the dashboard's engine.
DAX	Data Analysis Expressions – formula language used in Power BI to create logic trees and filters.
ALM	Asset Life Cycle Management.
WHPT	Wellhead Platform Tower

APM Asset Performance Management

Upstream 2.0 ADNOC's digital transformation vision for upstream oil and gas operations.

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